UNIVERSITI TEKNOLOGI MALAYSIA

Faculty of Computing

SECJ 3483 – WEB TECHNOLOGY

INDIVIDUAL ASSIGNMENT (20%)

Assignment Title : Web Application & Robotic Process Automation (RPA)

Mode : Individual

Duration : 6 Weeks

Weightage : 20% of Final Assessment

Event	Date
Challenge Launch	31st March 2026
Submission Deadline	8 May 2026
Assignment Evaluation Week	10 May – 26 May 2026
15 Shortlisted Assignment Review by DHL	27 May – 29 May 2026
Announcement of Top 10 Shortlisted Assignments	29 May 2026
Presentation in Imazium	3 June 2026

1. INTRODUCTION

Industry Collaboration & DAC 3.0 Challenge

This assignment is conducted in collaboration with a real industry partner, DHL Global Finance Services – Asia Pacific, through the DAC 3.0 Challenge.

Students will work on real-world logistics problems inspired by actual industry practices and requirements.

Based on the evaluation results:

- The top 10 projects from UTM JB and 5 projects from MJIIT will be shortlisted for the next stage of the DAC 3.0 Challenge.
- Shortlisted students will represent the faculty in the subsequent phase of the competition.

The overall winners of the DAC 3.0 Challenge will receive:

- Cash prizes
- Certificate of participation
- Internship opportunities at DHL Asia Pacific Shared Services under the Digital Automation Center Team

Participation in this assignment therefore provides not only academic assessment, but also real industry exposure and potential career opportunities.

*** Note:** *Selection for the DAC 3.0 Challenge is based on industry evaluation and is independent of the academic grading process.*

This assignment is weighted as follows:

- 60% – Web Application Implementation
- 40% – RPA Design & Automation Thinking

2. LEARNING OUTCOME

Upon completion of this assignment, students should be able to:

1. Develop an interactive web application using JavaScript and modern web technologies
2. Use JSON as a data format and consume RESTful APIs
3. Implement CRUD operations using a Mock JSON API server
4. Design a realistic Knowledge Base workflow
5. Manage Knowledge Base data effectively
6. Identify processes suitable for RPA automation
7. Explain automation logic using diagrams and demonstrations

3. LOGISTICS SCENARIO SELECTION

Scenario 1: AI-Powered Knowledge Base Automation for DHL Logistics Operations

DHL Logistics Operations teams handle thousands of messages and documents every day. These come from many different sources, such as:

- Chat messages from MS Teams or Telegram
- Long email threads between teams
- Screenshots used to explain steps or errors
- Handwritten notes or quick instructions shared informally
- Snippets of PowerPoint slides or training materials

All this information is important — but it is unstructured, scattered, and difficult to reuse. Because of this, creating clean and standardized SOPs (Standard Operating Procedures) or Knowledge Base (KB) articles becomes extremely slow and inconsistent.

Your Objective:

You are required to design and build a system that automates the entire transformation process — from raw messy input to clean knowledge articles.

Scenario 2: AI-Enhanced Incident Reporting & Resolution System

DHL's Customer Support teams receive a high volume of incident reports every day — late deliveries, address issues, damaged parcels, system errors, and customer complaints. These reports come through multiple inconsistent channels, such as:

- Email inboxes
- Telegram/Teams chat messages
- Phone call notes
- Images/screenshots of damaged packages
- Handwritten instructions from warehouse teams

Because the information is unstructured, incomplete, and inconsistent, it is difficult for DHL to:

- Identify the real issue
- Assign incidents to the correct department
- Prioritize incidents correctly
- Track resolution progress
- Produce consolidated reports for management

This leads to slow response times, repeated manual work, and inconsistent customer service quality.

Your Objective:

You are required to design and develop an AI-powered Incident Reporting System that automates the full process of collecting, understanding, assigning, and tracking incident reports.

4. SYSTEM REQUIREMENTS

4.1 Web Frontend – Web Application (Mandatory)

- Developed using any web technology (example: HTML, CSS, JavaScript, React, .NET)
- Interactive user interface with event handling
- Use JavaScript functions to manage system logic
- Provide form input and validation
- User-friendly navigation and interface design

4.2 Backend – JSON & Mock API OR Database (Mandatory)

Option 1:

- Use JSON format for data storage
- Use a mock API server such as JSON Server
- Implement CRUD operations (GET, POST, PUT/PATCH, DELETE)
- All data must be retrieved via API (no hardcoded data)

Option 2:

- All data handled via database storage (example: SQL, MongoDB)

4.3 RPA Component (Mandatory)

- Use UiPath Studio to create an RPA workflow
- Design an RPA workflow diagram
- Explain automation logic clearly

5. FUNCTIONAL REQUIREMENTS

5.1 Web Application (Mandatory)

- Secured access to the website, allowing RPA to create content
- Upload Console: accepts multiple input types (minimum: text, PDF and .docx as input)
- Draft and save information with status
- Viewer Page: searchable, filterable (by tag, date, creator, status)
- Versioning: show Draft → Reviewed → Published status with history
- Store creator details: at least basic login for editors/reviewers

5.2 RPA Automation (Mandatory)

- Ingestion: Read new files from Google Drive (or a designated email inbox exported to Drive)
- Duplicate checks: Skip items seen in the last 14 days (hash of text or file)
- Create new content in a web application and be able to attach files/screens
- Update the status in the web application
- Error Handling: Try/Catch + Take Screenshot on failures; write logs
- Send Summary Email to system admin when executed: totals for created, updated, duplicates, and failed; attach logs

5.3 Non-Functional Expectations (Optional)

- Integrate the usage of GPT (LLM) to summarize, structure, title, tag, and quality-polish articles; flag conflicts/outdated content; propose step-by-step procedures
- Draft Builder in Web Application: display AI-proposed Title, Summary, Steps, Tags, Related Links; editors can revise and save
- Conflict/Outdated Information Alerts in Web Application: flag if the new draft conflicts with existing published articles
- Read and transform information from images (PNG/JPG) via GPT as source data for knowledge base update in web application

6. DELIVERABLES

A. Source Code

- Web application project
- db.json dataset (if applicable)
- UiPath Project File
- Step-by-Step Instructions to run the system (Word document)

B. Report (8–12 pages)

- Introduction and scenario description
- System architecture
- API and JSON structure

- User interface screenshots
- RPA workflow and automation explanation

C. Demo Video (5–10 minutes)

- End-to-end system walkthrough with demo of functionality
- Demonstration of CRUD operations via API
- Explanation of RPA automation design

7. MARKING SCHEME (20 MARKS)

Component	Marks
A. Web Application & API Implementation • Knowledge Base Workflow & System Logic: 3 • User Interface & Interaction: 3 • JavaScript Functionality: 2 • Data Management: 2 • JSON & API Integration: 2	**12**
B. RPA Design & Automation • RPA Process Identification & Justification: 2 • RPA Workflow Design: 2 • Automation Logic & Trigger: 2 • RPA Integration & Explanation: 2	**8**
TOTAL	**20**

8. TIMELINE

Week	Activity
Week 1	Planning
Week 2	RPA Workflow & Logic
Week 3	Web Application – Setup & Routing
Week 4	API CRUD Implementation
Week 5	Integration & Testing
Week 6	Final Submission

9. ACADEMIC INTEGRITY

- This is an individual assignment. Collaborative submission is strictly prohibited.
- Plagiarism in any form is strictly prohibited and will be subject to disciplinary action.
- Students may be asked to explain their work during evaluation.

10. SUBMISSION INSTRUCTIONS

Submit ONE ZIP file containing the following items:

- Project Source Code (Web Application & RPA)
- db.json

- Report
- Video link

File name format:

`WT_Assignment_<MatricNo>_<Name>.zip`

APPENDIX A: MARKING RUBRIC

Subject	Web Technology
Assignment	Individual Assignment – Web App + JSON API + RPA
Total Marks	20
Rubric Scale	4 = Excellent 3 = Good 2 = Satisfactory 1 = Minimal

A. Web Application & API Implementation (12 Marks)

No.	Criteria	4: Excellent	3: Good	2: Satisfactory	1: Minimal	Weight	Score
A1	**Logistics Workflow & Business Logic**	Realistic logistics workflow with clear rules and status transitions	Workflow implemented correctly	Basic workflow implemented; logic is limited or partially complete	Very minimal or unrealistic workflow; limited functionality	**3**	
A2	**User Interface & Interaction**	Clear navigation and user feedback	Usable interface	Basic interface implemented	Poor interface	**3**	
A3	**JavaScript Functionality**	Structured functions and correct event handling	Mostly correct logic	Basic JS functionality	Incorrect or minimal JS	**2**	
A4	**Data Management**	Efficient data organisation with search/filter/sort	Data handled correctly	Basic data display	Inconsistent handling	**2**	
A5	**JSON & API Integration**	Correct JSON structure with full CRUD via API	Most CRUD operations working	Partial CRUD functionality	Minimal API usage	**2**	

B. RPA Design & Automation (8 Marks)

No.	Criteria	4: Excellent	3: Good	2: Satisfactory	1: Minimal	Weight	Score
B1	**RPA Process Identification & Justification**	Clearly identifies suitable automation processes with strong justification; aligns perfectly with logistics knowledge-base scenario; shows deep understanding of RPA applicability	Identifies relevant processes with acceptable justification; shows good understanding	Basic identification with limited reasoning; justification is shallow or generic	Poor or unclear process identification; justification missing or incorrect	**2**	

B2	**RPA Workflow Design**	Detailed, structured, and fully accurate workflow diagram; includes branching, decisions, exception paths, retry logic, and process governance outputs	Mostly correct workflow with clear structure; minor gaps in exception or branching logic	Basic workflow with partial correctness; lacks depth, limited to linear flow	Minimal or incorrect workflow; very limited understanding of RPA design	**2**	
B3	**Automation Logic & Trigger Design**	Automation logic is well-defined. Demonstrates correct step-by-step actions including ingestion, duplicate checks, web update, error handling, and summaries	Good logic with clearly described triggers; minor omissions in certain automation steps	Basic logic explained; trigger mechanism partially defined; lacks details on exception handling	Minimal logic described; trigger unclear; major missing steps	**2**	
B4	**RPA Integration & Explanation**	Excellent explanation of how RPA integrates with the web application (CRUD actions, API calls, data update, status transition); shows strong understanding of UiPath components	Good explanation of integration; minor missing details	Basic integration described; lacks clarity or depth	Minimal or unclear explanation; fails to show integration understanding	**2**	

APPENDIX B: EVALUATION FORM – INDIVIDUAL ASSIGNMENT

Subject	Web Technology
Assignment	Web App + JSON API + RPA
Total Marks	20

Student Name: ______________________ **Matric No.:** ______________________

Examiner: ______________________ **Date:** ______________________

A. Web Application & API Implementation (12 Marks)

Criteria	Weight	Score	Remarks
A1. Web Functionality & Logistics Workflow	5		
A2. Vue CLI Structure & Components	4		
A3. JSON + Mock API CRUD	3		

B. RPA Design & Automation (8 Marks)

Criteria	Weight	Score	Remarks
B1. RPA Process Identification & Justification	2		
B2. RPA Workflow Design	2		
B3. Automation Logic & Trigger Design	2		
B4. RPA Integration & Explanation	2		

TOTAL SCORE: _________ / 20

General Comments:

Examiner Signature: ______________________________